



Structure regulation of lignin-derived N-doped carbon-supported Ni catalyst for efficient upgrading of ethanol to higher alcohols

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ABSTRACT

The selective conversion of biomass-derived ethanol into higher alcohols is a desirable but challenging process for the efficient production of value-added chemicals from alternative carbon sources. However, the current catalysts used in ethanol coupling exhibit low conversion rates and yields towards the formation of higher alcohols, necessitating significant improvements in their efficiency for practical applications. In this work, a lignin-derived N-doped carbon-supported Ni catalyst for ethanol coupling was designed and synthesized. The Ni@NC catalyst, synthesized using diethylenetriamine as the nitrogen source with a mass ratio of nitrogen source to lignin of 2:1 and a molar ratio of metal to lignin of 20:1, demonstrated excellent performance, with an ethanol conversion rate of 81.1% and a yield of C₄₊ alcohols reaching 35.3%. Experimental results and density functional theory (DFT) calculations suggest that nitrogen-doped lignin, in appropriate quantities, along with the metal active component, forms a Ni-N-C structure that controls charge redistribution between the support and the Ni catalyst. This structure hinders the interaction of reaction intermediates and inhibits the production of by-products in the decomposition reaction, resulting in a higher yield of C₄₊ alcohols. This study presents a more cost-effective and efficient approach for developing ethanol-coupling catalysts and holds promise for expanding the utility of lignin-derived catalysts.

1. Introduction

The production of biofuels and chemicals from renewable, carbon-neutral biomass is gaining popularity as a feasible strategy to reduce greenhouse gas emissions [1–4]. Bioethanol, currently under development as a viable substitute or fuel additive for gasoline [5,6], is a key focus in this endeavor. Research on fermenting lignocellulosic biomass into ethanol has rapidly progressed over the past five years [7–10]. However, ethanol presents significant drawbacks when blended with gasoline, including poor energy density, hydrophilicity, and engine corrosion [11–13]. Advanced alcohols, such as butanol, exhibit fuel properties akin to gasoline and offer potential solutions to many of the challenges associated with ethanol [14–16]. Nonetheless, developing alternative technologies for efficiently converting ethanol to higher alcohols remains a significant challenge.

The rational design of solid catalysts is currently under consideration to achieve the significant scientific goal of selectively converting ethanol to higher alcohols. Among the most effective catalysts employed in ethanol coupling tests are homogeneous iridium and ruthenium complex catalysts [17,18], hydroxyapatite catalysts [19], and metal-supported heterogeneous catalysts such as Ru, Ir, Pd, and Ni [20,21]. However, they also exhibit drawbacks, including poor catalytic activity and harsh reaction conditions, resulting in limited ethanol conversion and an increased formation of gas byproducts [22]. Extensive research efforts have addressed this issue. For example, Shao et al. [23] achieved a maximum yield of 81.1% for a target alkane by preparing Ni/Mg-Al-O/AC loaded with 20wt.% Ni. Wu et al. [24] developed the Ni/(4-Mg)AlO catalyst based on hydrotalcite laminar plates, which achieved a remarkable 25.2% yield of C₆₊ alcohol and a 67.7% conversion of ethanol at 220 °C. In contrast, carbon materials derived from biomass

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sources, such as lignin, have garnered significant attention as low-cost and readily available carriers. Lignin, the second-largest natural organic polymer after cellulose [25,26], serves as an excellent precursor to carbon compounds due to its abundance in phenolic hydroxyls, alcohol hydroxyls, carboxyls, and other groups, with a carbon content of up to 60% [27–31]. These abundant functional groups effectively coordinate with metals to form a lignin-metal framework, thereby promoting the dispersion of metals. Through further in-situ reduction and carbonization, the metal catalyst is coated and embedded in the carbon carrier is formed, thus utilizing the synergistic effect between the metal and carbon layers to enhance catalytic activity and stability [32–35]. Additionally, lignin can adjust its properties through nitrogen doping to generate pyridinic-N and graphitic-N, while the rich defect structure further enhances the dispersion of active metals and improve the interaction with carbon supports [30,36,37]. Consequently, the lignin-derived nitrogen-doped carbon Ni catalyst exhibits superior catalytic performance.

In this study, we describe the preparation of a precursor for lignin-metal coordination composites using aminated lignin as a carrier for metal ions and nitrogen doping. Lignin-derived nitrogen-doped Ni-based carbon catalysts (Ni@NC) were synthesized via in-situ reduction and calcination. Under optimal conditions, ethanol conversion reached 81.1%, with a corresponding C_{4+} alcohol yield of 35.3%. Structural characterization of the catalysts, coupled with performance analysis and density functional theory (DFT) calculations, demonstrated that the precise anchoring of metal ions by nitrogen doping could enhance the defective structure of the catalysts. This enhancement facilitates the modulation of catalytic sites, improve the atomic utilization of the active components, and enables efficient catalytic conversion of ethanol to higher alcohols. This green synthesis strategy presents a promising and sustainable alternative biochar catalyst for the production of higher alcohols from ethanol.

2. Experimental

2.1. Materials

Alkali lignin (AL) with an average molecular weight of 6400 g/mol was procured from Shandong Longlive Biotechnology Co., Ltd. Ni ($(NO_3)_2 \cdot 6H_2O$ (98%), NaOH (97%), formaldehyde (37wt.%), ethanol ($\geq 99.5\%$), diethylenetriamine ($> 99\%$ GC), dimethyl sulfoxide (99%), DL-histidine, and n-amyl alcohol ($> 99\%$) were procured from Aladdin. Diethylenetriamine (99%), monochloroacetic acid (98%), and polyvinyl amine (M.W. 1800, 99%) were supplied by the Shanghai Maclin Reagent Company. The substances were used as received without further purification.

2.2. Preparation of Ni@NC catalysts

The modified lignin precursor was prepared through carboxylation and amination. The resulting samples are designated as X-AL-Z, where X represents the mass ratio of the aminating agent to the lignin ($X = 0, 1/4, 1/2, 3/4, 1$), and Z denotes the class of aminating reagent utilized (D for diethylenetriamine, P for polyvinyl amine, E for diethylenetriamine, and H for histidine). Additional synthesis details are provided in the [supporting information](#).

The preparation of the lignin-derived monometallic atomic catalyst, Ni@NC, via solvent volatilization, self-assembly, and calcination is depicted in Fig. 1. Nickel nitrate hexahydrate (1.84 g), dissolved in 5 mL of ultrapure water, was added dropwise to the modified lignin solution while stirring. Metal-lignin precursors with varying metal-to-lignin molar ratios were produced and designated as AL-Ni. The lignin-metal precursor was dried and crushed into a powder. AL-Ni (3 g) was placed into a tube furnace and heated to 500 °C in a N_2 atmosphere for 2 h pyrolysis using a specified heating process. The end product was a black solid Ni@NC that spontaneously cooled to ambient temperature. To expedite the process in subsequent studies, the catalyst was ground and designated as X-Ni_Y@NC-Z (where X denotes the mass ratio of the aminating agent to the lignin, Y denotes the molar ratio of the metal to the lignin, and Z denotes the class of aminating reagent utilized). Specific preparation details are described in the [supporting information](#).

2.3. Catalyst characterization

Fourier transform infrared (FTIR) spectroscopy was conducted using an iS50R all-band research-grade infrared spectrum workstation from Thermo Fisher to identify the functional groups present in the materials. The crystal structures of the catalysts were analyzed using a TD-3500 X-ray diffractometer. The surface microstructure was examined using high-resolution field-emission scanning electron microscopy (SEM, Zeiss Sigma 500). Transmission electron microscopy (TEM) was performed on an FEI Tecnai G2 F20 S-Twin instrument to further magnify the microstructure of the sample materials and observe the appearance and dispersion of the nanoparticles. Nitrogen adsorption and desorption isotherms were obtained using a Micromeritics ASAP 2020 instrument at 77 K. The N_2 adsorption/desorption isotherms and pore size distribution curves of the samples were generated by processing the data using Origin 2021 software to better understand the specific surface area and pore volume of the materials. Raman spectroscopy was utilized to assess the degree of carbon catalyst defects and graphitization. X-ray photoelectron spectroscopy (XPS) was employed to investigate the valence states and elemental composition of the catalyst using a Thermo Alpha + photoelectron spectrometer.

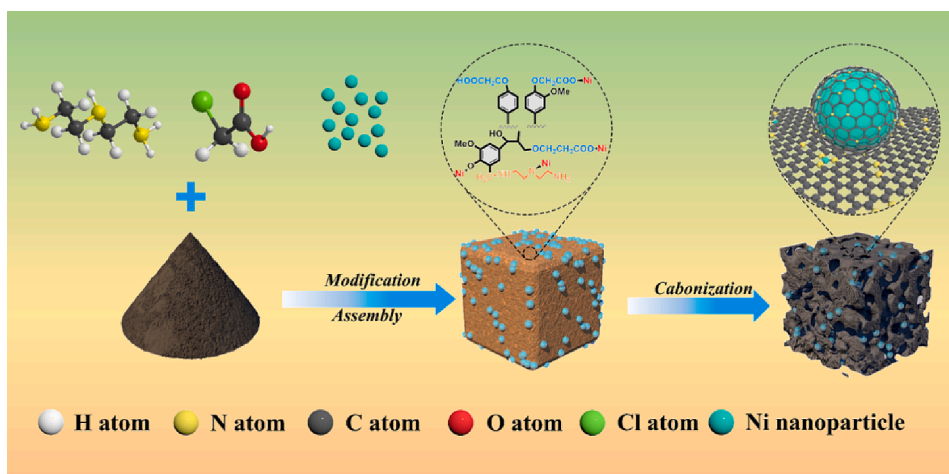


Fig. 1. Preparation diagram of Ni@NC catalyst.

2.4. Catalytic evaluation

Based on previous studies [20,21,28,38,39] and in conjunction with the results of control experiments (Table S1), the ethanol coupling reaction was conducted. Catalyst performance was evaluated through ethanol coupling studies in a high-pressure reactor (MS-50-316L, Anhui Kemi Machinery Technology Co., Ltd., Hefei, China). Initially, the reactor was loaded with 15 g of anhydrous ethanol, 15 g of ultra-pure water, 1 g of sodium hydroxide, and 0.5 g of catalyst. Air in the reactor was purged with hydrogen to create a hydrogen environment conducive to the reaction. The mixture was then heated to 250 °C and allowed to react for 24 h. Stirring was maintained at 500 rpm throughout the reaction.

2.5. Product analysis

Gas chromatography with a flame ionization detector (FID) and thermal conductivity detector (TCD) (GC 9800, Kechuang, China) was used to examine the gas composition after measuring the volume of the produced gaseous product. The liquid products were centrifuged, split into aqueous- and organic-phase samples, and weighed to determine the carbon balance during the reaction. Ethanol conversion, selectivity, and yield of the generated gaseous and liquid products were examined, with specific instrument characteristics, product analysis techniques, and calculation methodologies presented in detail in the [supplemental information](#).

2.6. DFT calculations

All calculations were performed using plane-wave-based density functional theory (DFT) as implemented in the Vienna Ab initio Simulation Package (VASP) [40,41]. A generalized gradient approximation (GGA) with the PBE exchange–correlation function developed by Perdew, Burke, and Ernzerhof was employed [42]. The DFT-D3 scheme developed by Grimme was used to account for the long-distance van der Waals (vdW) interactions [43]. All the atomic positions were fully relaxed using the conjugate gradient method. To investigate the influence of the N content within the support on the reaction, four support

models were constructed: one graphene carbon support (C) and three N-doped carbon supports (C_xN) with different C/N ratios of 3, 8, and 11. A 6×6 unit cell with 72 atoms was used to simulate the support, and a Ni_{10} cluster model was adopted to simulate the Ni catalyst. A $2 \times 2 \times 1$ Monkhorst–Pack grid centered on the γ point was utilized for the K-point sampling. A vacuum region of 15 Å was introduced perpendicular to the surface direction to eliminate interactions between neighboring layers. The convergence criteria of force and energy were set as 0.02 eV/Å and 10^{-5} eV, respectively.

3. Results and discussion

3.1. Catalyst morphological and structural analysis

To analyze the coordination complexation of lignin and metal ions, the FTIR profiles of lignin and metal-lignin precursors modified with various quantities of aminating agents are presented in Fig. 2a. The spectrum of alkali lignin (Fig. S1a) exhibits a strong absorption band in the 3700–3200 cm^{-1} range, representing the stretching vibration peaks of phenol and alcohol hydroxyl groups. Absorption peaks at 1715–1660 cm^{-1} indicate the presence of C=O groups, while peaks at 1610–1600, 1520–1500, and 1450–1420 cm^{-1} represent the skeletal vibration of the benzene ring.

Comparison of the FTIR spectra of modified lignin and alkaline lignin suggests that carboxymethylation modification effectively introduces –COOH groups, as evidenced by the enhanced absorption peak of the C=O bond at 1715–1660 cm^{-1} . The FTIR spectra of amine-modified and carboxymethylated lignin show a strengthened –NH shear vibration peak at 1650–1590 cm^{-1} and the emergence of a –CN absorption peak at 1135 cm^{-1} . Metal-lignin precursors modified by diethylenetriamine exhibit the strongest –CN absorption peak compared to those modified by other aminating agents (Fig. S1b), indicating successful grafting and the superior grafting effect of diethylenetriamine.

The absorption band at 1380 cm^{-1} significantly increases in the FTIR spectra of metal-lignin precursors and amine-modified lignin with varying amounts of aminating reagents, confirming the presence of the –NO₂ group. This is attributed the use of nickel nitrate throughout the experiment, as well as the effective grafting of Ni onto the modified

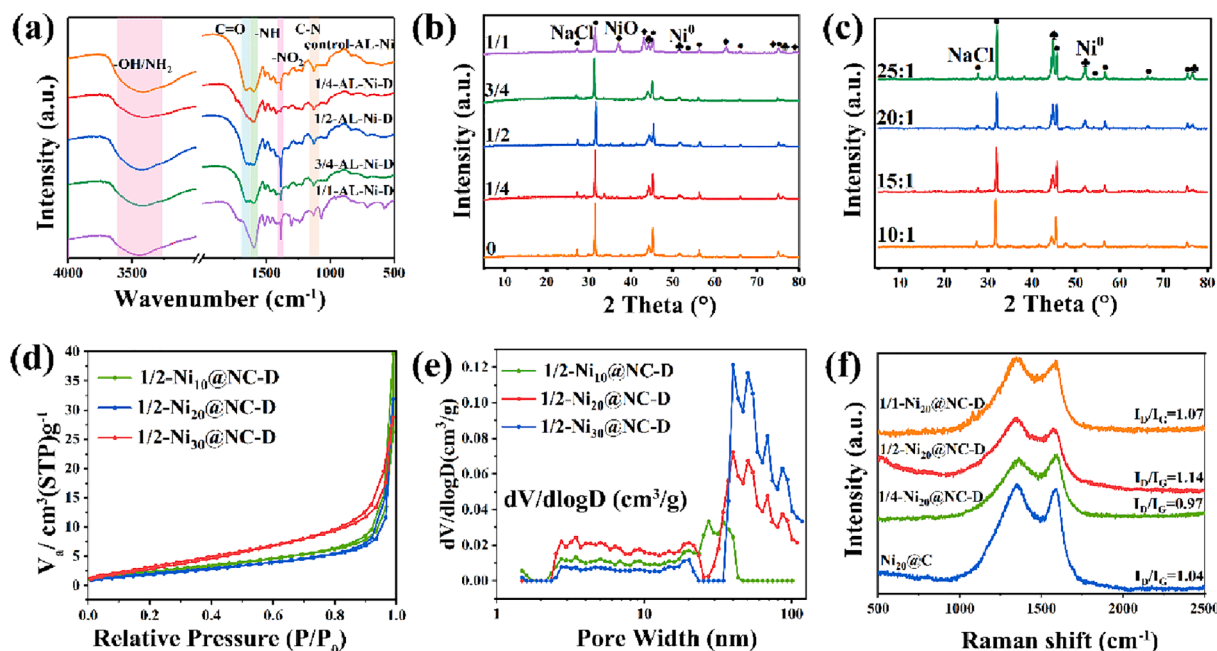


Fig. 2. (a) FTIR spectra of lignin modified with various degrees of amination and XRD pattern of 1/2-Ni@NC-D catalysts (b) with various lignin to metal molar ratios and (c) various amination reagent to lignin mass ratios. (d) Nitrogen adsorption – desorption isotherms and (e) pore size distributions of 1/2-Ni@NC-D catalysts. (f) Raman spectra of 1/1-Ni₂₀@NC-D, 1/2-Ni₂₀@NC-D, and Ni₂₀@C catalysts.

lignin, resulting in the formation of a $-\text{NO}_2$ absorption band [25,44,45].

The crystalline phase compositions of the Ni@NC catalysts with different metal-to-lignin and reagent-to-lignin mass ratios were determined by XRD (Fig. 2b–c). Typical peaks for NiO occur at $2\theta = 44.4^\circ$, 51.8° , and 76.6° , corresponding to the (1 1 1), (2 0 0), and (2 2 0) crystal planes of elemental nickel, respectively. When the mass ratio of the amination reagent to lignin increased to 1, the apparent peaks of the Ni₂₀@NC-D catalyst appeared at $2\theta = 37.2^\circ$, 43.3° , 62.9° , 75.4° , and 79.4° . With an increase in the amount of amination reagent, a portion of NiO underwent conversion to NiO. Notably, the intensity of the diffraction peaks of nickel increased as the metal-to-lignin ratio rose from 10:1 to 25:1. The diffraction pattern exhibits distinct and elongated peaks, indicating the highly crystalline structure of the catalyst.

The N₂ adsorption/desorption isotherms and pore size distributions of the 1/2-Ni@NC-D catalysts produced with various amounts of metal and lignin are illustrated in Fig. 2d and e. The nitrogen adsorption and desorption curves display a noticeable hysteresis loop in the P/P_0 range of 0.6–1.0, suggesting that the catalysts in this series are predominantly mesoporous with pore sizes ranging from 25–35 nm. Table 1 presents the specific surface area, pore volume, and average pore diameter of the mesoporous materials with different metal-to-lignin ratios (1/2-Ni@NC-D), as determined by the Brunauer-Emmett-Teller (BET), YANG, and Barret-Joyner-Halenda (BJH) equations. The results indicate that the pore size distribution range of the catalyst was rather broad, while the particle size was uniform. Moreover, nitrogen doping promotes the development of an abundant pore structure, uniform metal particle distribution, and reactant adsorption onto the catalyst, thereby improving the ethanol conversion rate.

Raman spectra, as depicted in Fig. 2f, were utilized to analyze how the ratio of the aminating agents affected the degree of defects and graphitization of the catalyst. Peaks D and G, situated at approximately 1350 cm^{-1} and 1580 cm^{-1} , respectively, represent Raman characteristic peaks of carbon atom crystals. The I_D/I_G ratio served to identify the carbon species. Fig. 2f illustrates the Raman spectra of 1/2-Ni₂₀@NC-D and Ni₂₀@C catalysts with a molar ratio of 20:1 metal to lignin, modified by 1/2 amination using diethylenetriamine as the aminating reagent. In contrast, the 1/2-Ni₂₀@NC-D and 1/1-Ni₂₀@NC-D catalysts exhibited I_D/I_G values of 1.14 and 1.07, respectively, larger than those of Ni₂₀@C (1.04). The degree of defect decreased as the mass ratio of the aminating reagent to lignin increased. Appropriate nitrogen doping enhanced the formation of defect structures. The catalyst prepared with a 1/2 amination ratio contains a significant amount of defect structure, affecting the catalytic performance of the catalysts and the conversion of ethanol to higher alcohols.

As shown in Fig. S2, carboxylated and aminated lignin enhanced the concentrations of N and O in the lignin derivatives. This modification was confirmed using X-ray photoelectron spectroscopy and elemental analysis. The microstructure of the 1/2-Ni₂₀@NC-D catalyst, subjected to 1/2 amination using diethylenetriamine as the aminating reagent, is illustrated in Fig. 3a and b. The catalyst exhibited a well-developed pore structure, interpenetration, lamellar structure, and uniform distribution

of metal particles, as confirmed by BET characterization. The XRD pattern reveals that the amorphous carbon coating on the crystal particles has lattice fringes with a spacing of $2.03\text{ \AA}$, corresponding to the (1 1 1) crystal face of Ni, thus providing further evidence of the presence of the active Ni⁰ component (Fig. 3c). As illustrated in Fig. 3d, metal particle sizes ranged between 6 and 22 nm, with an average particle size of 12.92 nm. No visible pores were observed in the Ni₂₀@C catalyst, while some metal particles were agglomerated (Fig. S3a and b), comprising particles ranging in size from 10–35 nm with an average particle size of 20.89 nm (Fig. S3c). According to the HAADF-STEM atlas (Fig. 3f) and the EDS elemental atlas (Fig. 3g–i), Ni is evenly dispersed on the carbon materials, owing to its strong coordination with amine groups. The Ni concentration measured by ICP was 22.42%, roughly consistent with the doping quantity.

XPS was employed to further analyze the chemical compositions and valence states of the catalysts. The high-resolution XPS spectra of the 1/2-Ni₂₀@NC-D and Ni₂₀@C catalysts were tested, as shown in Figs. 4 and S4a–c, respectively. Ni 2p, N 1s, and C 1s signals were detected in the full-scan spectrum (Fig. 4a), confirming the presence of Ni, N, and C elements on the catalyst's surface. These observations are further supported by peaks in the C 1s spectrum of 1/2-Ni₂₀@NC-D for C/C = C (284.6 eV), C-N (285.4 eV), C = O (288.1 eV), CO_3^{2-} (289.9 eV), and C-O (286.5 eV). The appearance of the CO_3^{2-} peaks may be attributed to the reaction of NaOH and its byproduct CO_2 with Na_2CO_3 . Conversely, the Ni₂₀@C catalyst exhibits only partial peaks for C/C = C, C = O, and CO_3^{2-} .

The N 1s spectrum of the 1/2-Ni₂₀@NC-D catalyst displays peaks corresponding to pyridine nitrogen (398.05 eV), N-Ni (398.3 eV), pyrrole nitrogen (399.9 eV), and graphitic nitrogen (400.3 eV). In contrast, the Ni₂₀@C catalyst exhibits only partial N-Ni and pyrrole nitrogen peaks. The XPS spectra of catalyst 1/2-Ni₂₀@NC-D reveal a C-N bond (285.05 eV), consistent with the C 1s spectra of the two catalysts, indicating effective nitrogen incorporation into carbon materials through amination, resulting in various forms of nitrogen through doping. Additionally, the presence of a Ni-N bond (855.8 eV) in the Ni 2p spectrum suggests successful coordination between the amino functional groups on lignin and Ni, leading to the even distribution of the active component Ni on the carbon carrier.

3.2. Catalytic performance analysis

The effect of modifying the 1/2-Ni₂₀@NC catalyst with various aminating reagents on ethanol coupling is depicted in Fig. 5. When the 1/2-Ni₂₀@NC-D catalyst was employed for catalyzing ethanol conversion to higher alcohols, the ethanol production rate and yield of the C₄₊ higher alcohols were 81.1% and 35.3%, respectively. The ethanol conversion rates of the 1/2-Ni₂₀@NC catalysts prepared using polyethyleneimine, diethylenediamine, and DL-histidine were 69.2%, 65.6%, and 54.8%, respectively, while the yields of the C₄₊ higher alcohols were 25.3%, 25.9%, and 22.3%. The carbon balance and gas product selectivity of the 1/2-Ni₂₀@NC-D catalyst are 76.7% and 32.4%, respectively. The carbon balances of the 1/2-Ni₂₀@NC catalysts produced with polyvinyl amine, diethylenediamine, and DL-histidine were 82.2%, 84.6%, and 81.1%, respectively, and the gas product selectivities were 40.4%, 35.7%, and 23.0%, respectively.

Based on the FTIR spectra of the 1/2-Ni₂₀@NC precursor synthesized with various aminating reagents (Fig. S1b), the absorption peak of $-\text{CN}$ at 1135 cm^{-1} was most prominent in the diethylenetriamine-modified precursor, further supporting the notion that diethylenetriamine, when used as an aminating agent, facilitates lignin amination through the Mannich reaction, enabling better complexation with Ni and enhancing catalytic efficiency. The XRD patterns of catalysts prepared with different amination reagents are illustrated in Fig. S1c. All catalysts exhibit sharp diffraction peaks, indicating that a highly crystalline structure. They all feature diffraction peaks representing Ni⁰. Compared with the 1/2-Ni₂₀@NC-D catalyst, the diffraction peaks of the 1/2-

Table 1

Structural characteristics of 1/2-Ni@NC-D catalysts produced in various metal-to-lignin ratios.

Catalyst.	S_{BET} (m^2/g) ^a	V_{Total} (cm^3/g) ^a	D (nm) ^a	Ni Surface Area ($\text{m}^2/\text{g}_{\text{Ni}}$) ^b	Ni content (wt.%) ^c
1/2-Ni ₁₀ @NC-D	9.09	0.0614	27.02	0.243	12.51
1/2-Ni ₂₀ @NC-D	12.72	0.0447	14.06	0.289	22.42
1/2-Ni ₃₀ @NC-D	7.71	0.0494	25.63	0.545	33.79

^a Determined by N₂ adsorption–desorption, ^b Determined by H₂ pulse chemisorption, ^c Determined by ICP-OES.

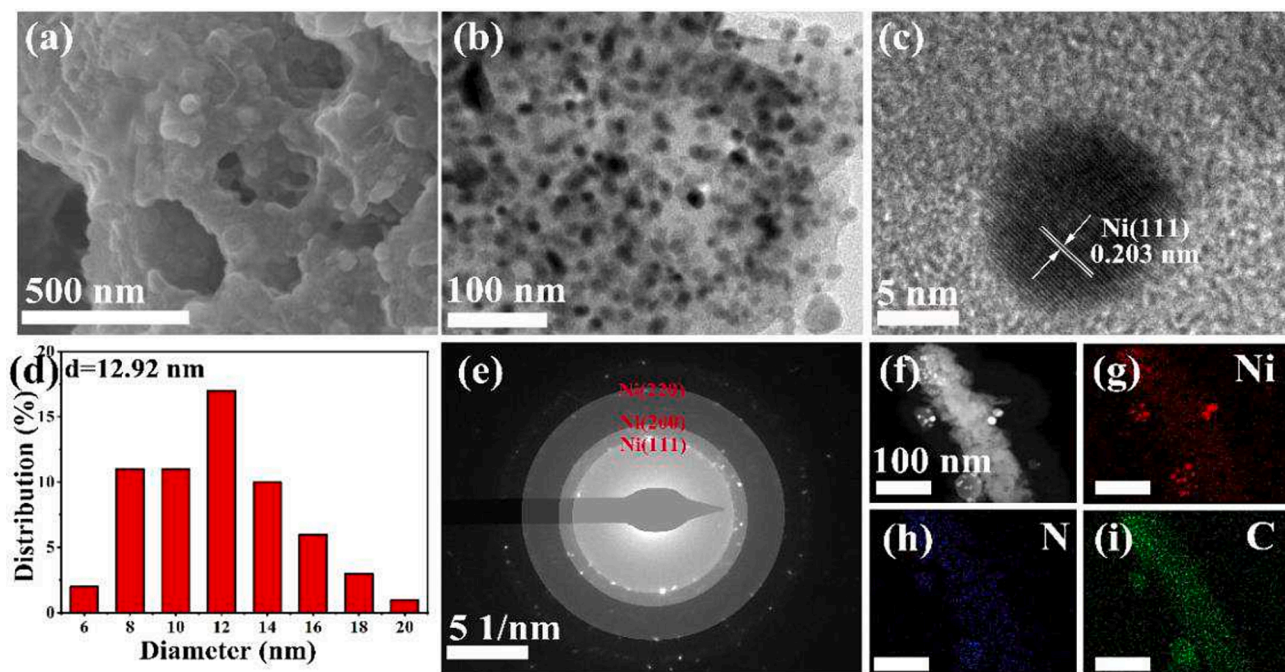


Fig. 3. (a) SEM image, (b) TEM image, (c) HRTEM image, (d) distribution of particle sizes, (e) SAED image, (f) HAADF-STEM image, and (g-i) EDS elemental map of Ni, N, C for 1/2-Ni₂₀@NC-D catalyst.

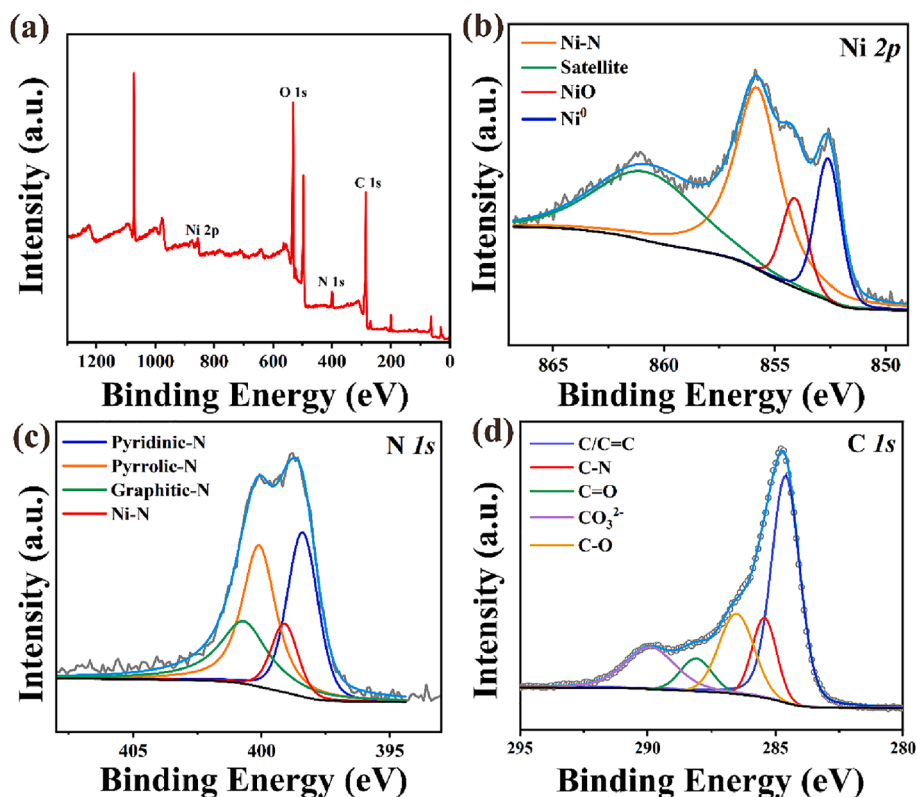


Fig. 4. XPS spectra of catalyst 1/2-Ni₂₀@NC-D: (a) XPS survey, (b) Ni 2p spectrum, (c) N 1s spectrum, and (d) C 1s spectrum.

Ni₂₀@NC catalyst modified with other nitrogen sources are relatively higher, and the bottoms are narrower, indicating higher crystallinity and larger particle sizes for these three catalysts. Consequently, the active sites are less exposed, leading to decreased catalytic efficiency. The distribution of carbon in the higher alcohols is illustrated in Fig. 5b,

where n-butanol is the primary product. However, the yields of higher alcohols decreased significantly as the number of carbon atoms increased.

The effects of Ni₂₀@NC catalysts prepared with various degrees of amination modification are depicted in Fig. 6a. The Ni₂₀@C catalyst,

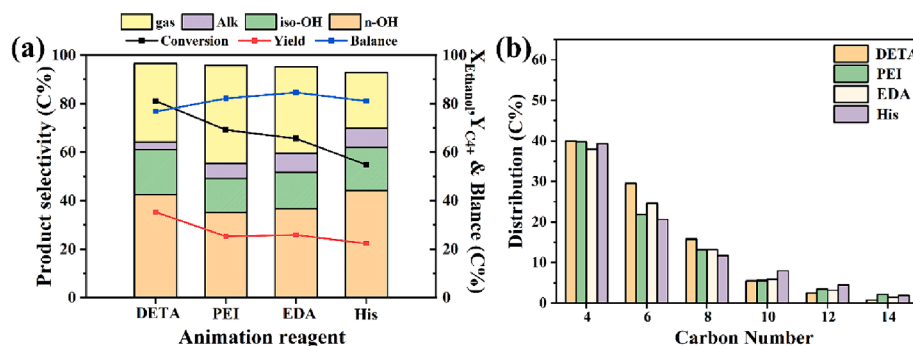


Fig. 5. Effect of 1/2- Ni_{20} @NC catalysts modified with different aminating reagents on the conversion of ethanol to higher alcohols: (a) ethanol conversion, higher alcohol yield, product selectivity, and carbon balance, (b) carbon distribution of higher alcohol products.

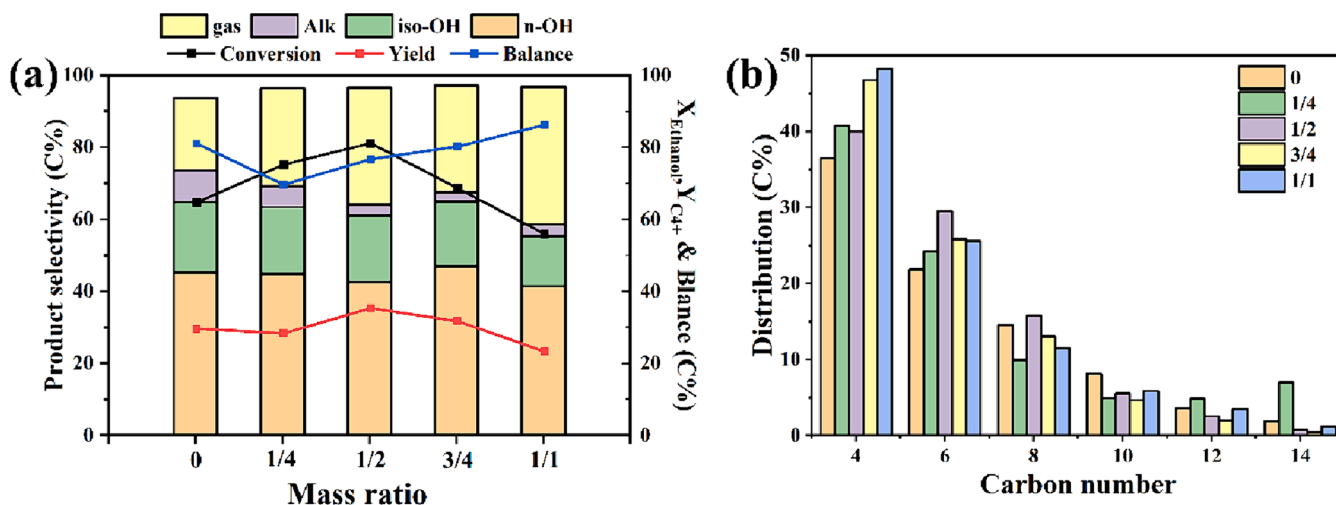


Fig. 6. Effect of Ni_{20} @NC-D catalysts prepared from modified lignin with various degrees of amination on the conversion of ethanol to higher alcohols: (a) ethanol conversion, higher alcohols yield, product selectivity, and carbon balance, (b) carbon distribution of higher alcohol products.

prepared from carboxymethyl lignin without amination, exhibited an ethanol conversion of 64.7% and a yield of 29.6% for high alcohols. The conversion rate of ethanol and the yield of C_{4+} higher alcohols increased from 75.2% to 81.1% and 28.3% to 35.3%, respectively, as the mass ratio of the amination reagent to lignin was increased from 1/4 to 1/2. However, as the mass ratio of the amination reagent to lignin continued to increase to 1/1, the conversion rate of ethanol and the yield of C_{4+} higher alcohols decreased to 55.9% and 23.3%, respectively. The carbon distribution in the higher alcohols is illustrated in Fig. 6b, where the major product was n-butanol, and the yield of C_{4+} alcohols decreased

with increasing carbon numbers.

The SEM image of the 1/1- Ni_{20} @NC-D (Fig. S5a) illustrates that the synthesized catalyst lacks an extensive pore structure. The TEM images and particle size analysis of the catalyst (Fig. S5b and c) reveal uneven metal particle distribution in the catalyst, with an average particle size of 17.37 nm, larger than the average particle size of the 1/2- Ni_{20} @NC-D catalyst (12.92 nm). Based on the aforementioned characterization data, an adequate nitrogen concentration promotes the anchoring of Ni^0 on carbon carriers and the creation of defect structures, resulting in an even distribution of the metal particles with higher dispersion and smaller

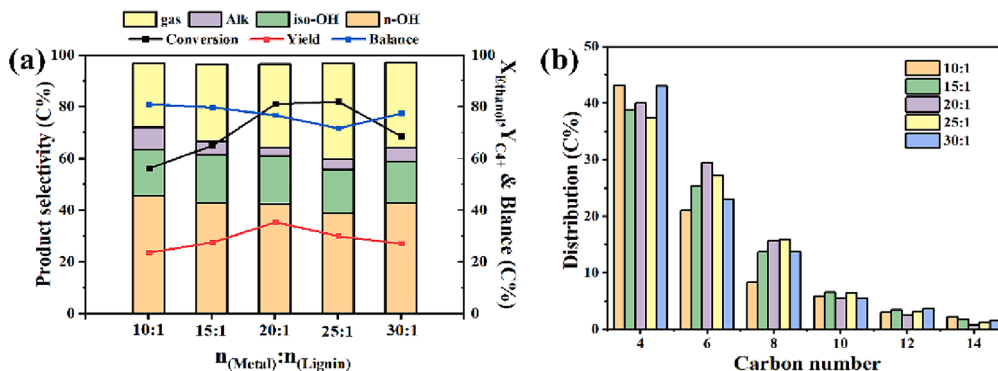


Fig. 7. Effect of 1/2- Ni @NC-D catalysts prepared from modified lignin with varied metal and lignin amounts on the conversion of ethanol to higher alcohols: (a) ethanol conversion, higher alcohols yield, product selectivity, and carbon balance, and (b) carbon distribution of higher alcohol products.

size. However, excessive nitrogen content diminishes the pore structure and defect structure of the carbon support and hampers the metal-support interaction, causing the agglomeration of metal ions and the partial conversion of Ni^0 to NiO , resulting in a significant decline in the catalytic activity of the catalyst.

Fig. 7a depicts the effect of varying metal and lignin amounts on catalytic ethanol coupling. When the metal-to-lignin molar ratio was 20:1, the ethanol conversion rate and yield of C_{4+} -higher alcohols were 81.1% and 35.3%, respectively, slightly higher than those of the 1/2-Ni@NC-D catalyst prepared with other metal-to-lignin ratios. Adjusting the metal-to-lignin molar ratio from 10:1 to 20:1 resulted in an increase in ethanol conversion rate from 56.2% to 81.9%, while the yield of C_{4+} higher alcohols improved dramatically from 23.6% to 35.3%. However, when the metal-to-lignin molar ratio was increased to 30:1, the production of higher alcohols decreased from 81.1% to 68.5%, with the yield of higher alcohols dropping from 35.3% to 27.0%. The carbon distributions of higher alcohols are shown in Fig. 7b, where the major product was n-butanol, and the yield of C_{4+} alcohols decreases with increasing carbon numbers. These findings indicate that the quantity of the metal catalyst is the primary factor affecting the catalytic activity of the catalyst when the molar ratio of metal to lignin is 10:1–20:1. Both the conversion of ethanol and the yield of higher alcohols dramatically increased with an increase in the metal-to-lignin molar ratio. However, because Ni has significant bond-breaking and dehydrogenation properties, the selectivity for gaseous byproducts increases as the metal-to-lignin molar ratio increases. The selectivity for gaseous byproducts increased from 24.8% to 36.9% when the molar ratio of metal to lignin increased from 10:1 to 25:1. In contrast to the 1/2-Ni₁₀@NC-D and 1/2-Ni₃₀@NC-D catalysts, the 1/2-Ni₂₀@NC-D catalyst exhibited an apparent porous structure, as indicated by the SEM spectra comparison (Fig. S6a and d). Catalyst 1/2-Ni₂₀@NC-D has the highest specific surface area (12.72 m^2/g), according to Table 1. With the addition of the active metal components from a metal-to-lignin molar ratio of 10:1 to 30:1, the active specific surface area increased from 0.243 $\text{m}^2/\text{g}_{\text{cat}}$ to 0.545 $\text{m}^2/\text{g}_{\text{cat}}$. However, when the molar ratio of metal to lignin was 30:1, agglomeration between the metals occurred (Fig. S6b and e), and the average particle size of the metal particles was 33.16 nm (Fig. S6f), significantly larger than the 12.92 nm for 1/2-Ni₂₀@NC-D and 13.90 nm (Fig. S6c) for 1/2-Ni₁₀@NC-D. This hindered the contact between active components and reaction substrates, resulting in a decrease in catalytic activity.

3.3. Recovery efficiency

The results of four ethanol coupling cycles performed on catalyst 1/2-Ni₂₀@NC-D are presented in Table 2. The ethanol conversion decreased from 81.1% to 43.8%, and the production of higher alcohols declined from 35.3% to 20.8%, although higher alcohol selectivity increased and gas byproduct selectivity decreased. Comparison of the XRD patterns (Fig. 8a) and SEM and TEM images (Fig. 8d–f) for the freshly prepared catalyst and the spent catalysts after one and four cycles indicated structural variations in the 1/2-Ni₂₀@NC-D catalyst, yet all maintained rich pore structures. N, C, and O are evenly dispersed on the carbon material, while the dispersion of Ni nanoparticles decreases. In the XPS spectra (Figs. 8b–c, S4d), the major states of Ni before and after the cyclic reaction remained Ni-N and Ni^{2+} . The comparison results

of XPS suggest that the cyclic reaction oxidizes the initially supported Ni^0 on the catalyst to Ni^{2+} , which then exists on the catalyst in the form of $\text{Ni}(\text{OH})_2$. Moreover, the binding energy of Ni shifts to a lower position, indicating a decrease in the electron density of Ni. Therefore, combined with previous studies and the above characterization results [46,47], the decrease in Ni electron cloud density may be attributed to the sintering phenomenon of nickel particles under high temperature and other reaction conditions. During the reaction process, carbon is deposited on the surface of the nickel catalyst, leading to the coverage of active sites and resulting in a reduction of surface active sites. The hydrogenation of the ethanol coupling process is further accelerated. The Ni content of the catalyst decreased as the number of cycles increases. Consistent with the results of ICP-OES, the Ni content decreased from 22.42% to 19.03%, resulting in a decrease in ethanol conversion and byproduct yield, such as gas. Therefore, the decrease in the catalytic performance of the catalyst may be attributed to changes in the pore structure of the catalyst during the reaction and recycling process, resulting in a reduction in the number and exposed area of active centers, which hinders effective contact between reactants and active sites. Moreover, a significant amount of mass loss occurs in the catalyst during the recovery process, which also contributes to a decrease in the ethanol conversion rate. However, the increase in alcohol product selectivity and the decrease in gas by-product selectivity may be attributed to the presence of Ni^0 on the surface of the fresh catalyst, which promotes the progression of side reactions. With the reaction and recycling process, Ni^0 on the catalyst surface is consumed or removed, leading to the suppression of side reactions.

3.4. Catalytic mechanism and discussion

The ethanol coupling on the Ni@NC catalyst for the preparation of higher alcohols follows the typical Guerbet reaction mechanism. It is widely accepted that the Guerbet reaction mechanism consists of three steps: dehydrogenation, aldol condensation, and hydrogenation. Initially, ethanol molecules undergo adsorption on the metal Ni site, leading to their dehydrogenation into intermediates such as $\text{CH}_3\text{CH}_2\text{O}^*$, CH_3CHO^* , CH_3CO^* , $\text{CH}_2\text{CH}_2\text{OH}^*$, etc. Subsequently, these intermediates are catalyzed by the NC supports possessing aldol condensation activity, resulting in the formation of $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{O}^*$. Upon the generation of $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{O}^*$, further dehydration and hydrogenation occur on the metal Ni site, yielding the desired product, n-butanol. However, it should be noted that the metal Ni also exhibits significant C-C/C-O bond cleavage activity, leading to C-C bond cleavage in the aforementioned intermediates and the production of by-products such as CH_4 , CO , CO_2 , etc., thereby reducing the selectivity towards high-carbon alcohols [39,48,49].

The impact of the N content in the NC support on the catalytic activity of the Ni nanoparticles was investigated across a range from 0 to 1/3. The optimized structures of Ni_{10}/C and $\text{Ni}_{10}/\text{C}_x\text{N}$ ($x = 11, 8$, and 3) are shown in Fig. 9a. As illustrated in Fig. 9a, the structure of Ni_{10} remained largely unchanged across different supports, exhibiting a two-layer geometry with five Ni atoms in each layer. Therefore, the different activities observed among the four model catalysts are attributable to differences in the N contents of the supports. The investigation into the reaction mechanism of ethanol coupling, as well as the side ethanol decomposition reactions, unveiled that the rate-determining steps encompass the C-C coupling of acetaldehyde with $\text{CH}_2\text{CH}_2\text{OH}$ and C-C bond cleavage, respectively, consistent with the propositions made in our prior research [39].

In this study, we conducted further investigation into these two rate-limiting steps. The optimized configurations of the relevant intermediates, as presented in Fig. 9b, were obtained on Ni_{10} supported by C_3N . It's worth noting that these configurations are representative of other supports as well, since the N/C ratio in the support does not affect the adsorption geometries. The reaction energies for the rate-determining C-C coupling and C-C cleavage reactions were found to be

Table 2
Reusability of the spent 1/2-Ni₂₀@NC-D Catalyst in aqueous ethanol upgrading.

Use cycle	Mass (g)	X_{Ethanol} (C%)	$Y_{\text{C}_{4+}}$ (C%)	Balance (C%)	Product selectivity (C%)	
					OH	Gas
1	0.50	81.1	35.3	76.7	61.0	32.4
2	0.26	65.5	31	73.2	79.9	13.5
3	0.15	56.0	29.5	78.4	85.9	6.8
4	0.09	43.8	20.8	85.6	70.8	12.1

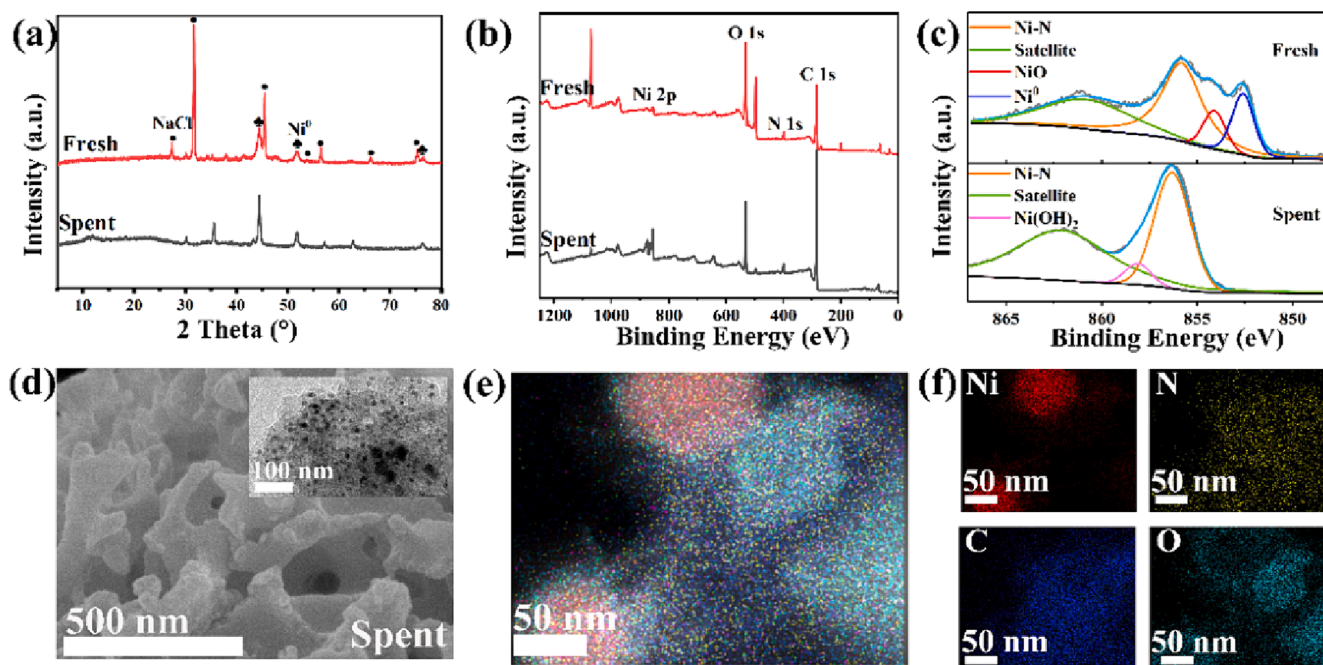


Fig. 8. (a,b) XRD, (c) XPS results of the fresh and spent 1/2- Ni₂₀@NC-D catalyst, (d) TEM, and (e, f) HAADF-STEM image and EDS elemental map of C, N, and Ni for the spent 1/2-Ni₂₀@NC-D catalyst.

−0.27 eV and −0.58 eV, respectively, on Ni₁₀/C (Fig. 9c). With increasing the N content in the NC support, the reaction energy of C-C coupling initially decreases to −0.32 eV and −0.36 eV at N/C ratios of 1/11 and 1/8, respectively, before increasing again to −0.15 eV at N/C ratio of 1/3. Similarly, the reaction energy of the side C-C cleavage reaction followed a similar trend, reaching the lowest value of −0.81 eV at an N/C ratio of 1/8, followed by an elevated value of −0.58 eV at an N/C ratio of 1/3. These findings suggest that the N content in the support affects the activity of Ni in a non-monotonic manner for both the ethanol coupling and side decomposition reactions, reaching maximum activity at intermediate N contents, which is consistent with the catalytic performance measurements shown in Fig. 4. As shown in Fig. 9d, the N content in the support significantly affected the charge redistribution between the Ni nanoparticles and the NC (or C) support.

On the basis of control experiments and DFT calculations, in conjunction with prior research, a plausible a reasonable mechanism for the production of higher alcohols from ethanol employing an appropriate amount of nitrogen-doped Ni@NC catalyst is proposed, as delineated in Fig. 9e. In the Guerbet reaction, curtailing the production of by-products by prohibiting the interaction between intermediates and the catalyst is paramount to enhancing the yield of C₄₊ alcohols. Studies have revealed that the quantity of N doping modulates the electronic structure of Ni [22,50–52], inducing a shift in intrinsic catalytic performance to augment the desorption behavior of intermediates such as CH₃CHO*, thereby promoting C-C coupling and mitigating C-C cleavage. With increasing N content in the support, the Ni₁₀ cluster assumes a less positively charged state. Given that the Ni₁₀ cluster exhibits a two-layered structure and the intermediates are adsorbed primarily on the top layer, the top five Ni atoms are more crucial to its adsorption properties. Based on the Bader charge analysis, there is a redistribution of charge between the support and Ni, resulting in electron accumulation on the top-layer Ni₅, which has charges of −0.12, −0.38, −0.31, and −0.28 for Ni₁₀/C, Ni₁₀/C₁₁N, Ni₁₀/C₈N, and Ni₁₀/C₃N, respectively. The N content in the support dramatically affects the charge redistribution between the support and the Ni catalyst, resulting in electron-rich Ni as the active sites at intermediate N contents. The charges of the top five Ni atoms also influenced the adsorption energy of the intermediates in a non-monotonic manner (Fig. S7), resulting in the lowest adsorption

strength for the coupling fragments and the highest ethanol coupling activity at a moderately negative charge (Ni₁₀/C₈N).

4. Conclusions

Using alkaline lignin as the carbon source, we employed amination modification engineering to determine the optimal nitrogen source and concentration. The experimental findings revealed that the 1/2-Ni₂₀@NC-D catalyst, synthesized with diethylenetriamine as the aminating reagent and a mass ratio of aminating reagent to lignin of 1/2, exhibited excellent performance. The ethanol conversion rate and alcohol output reached remarkable levels of 81.1% and 35.3%, respectively, underscoring the catalyst's outstanding catalytic efficiency. By incorporating an appropriate nitrogen doping content, the lignin catalyst effectively modulates the charge redistribution between the carrier and the Ni catalyst, thereby constructing a Ni-N-C structure with active metal components. This configuration results in electron-rich Ni sites serving as the active site at intermediate N content. Consequently, the adsorption strength of the moderately negative charge is minimized, leading to enhanced catalytic activity and promoting the synergistic effect between the catalyst and carbon carrier, ultimately boosting alcohol yield. The lignin-derived carbon-encapsulated Ni@NC catalyst, synthesized through amination modification engineering, holds immense potential for various applications in the catalytic conversion of higher alcohols. It is anticipated to broaden the scope of utilization for lignin-derived catalysts.

CCRediT authorship contribution statement

Quanzhou Xu: Writing – original draft, Investigation. **Xing Fei:** Writing – review & editing, Data curation. **Xueqing Qiu:** Writing – review & editing. **Xiaofei Wang:** Writing – review & editing, Formal analysis. **Tiejun Wang:** Writing – review & editing. **Xuliang Lin:** Writing – review & editing, Conceptualization. **Sha Li:** Writing – review & editing, Methodology. **Yanlin Qin:** Writing – review & editing, Supervision.

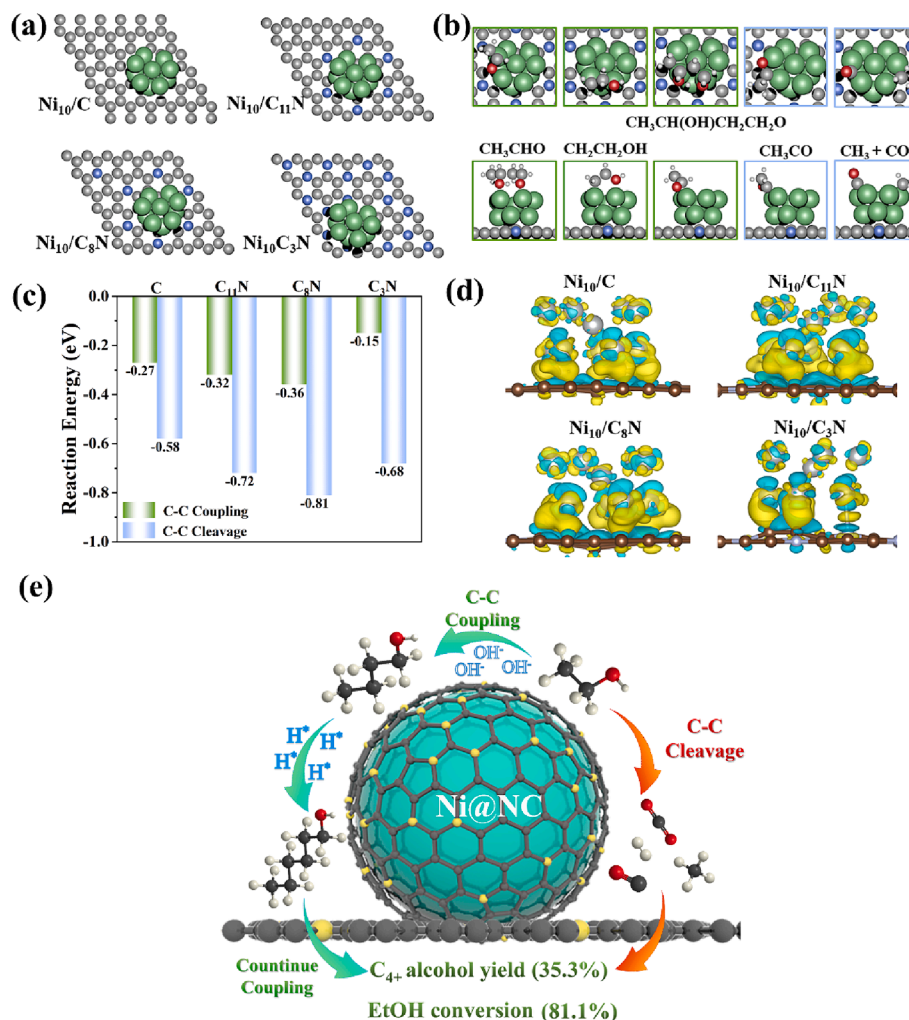


Fig. 9. (a) Optimized structures of Ni₁₀/C, Ni₁₀/C₁₁N, Ni₁₀/C₈N, and Ni₁₀/C₃N, (b) top view (top panel) and side view (bottom panel) of reaction intermediates on Ni₁₀/C₃N, (c) reaction energy of the rate-determining steps for ethanol coupling (C-C coupling) and ethanol decomposition (C-C cleavage), (d) differential charge density plots of Ni₁₀ supported on C, C₁₁N, C₈N and C₃N (yellow indicates electron accumulation and blue indicates electron deficiency), and (e) reaction pathway of ethanol to higher alcohols.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2024.151092>.

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